

MASARYKOVA
UNIVERZITA
PŘÍRODOVĚDECKÁ FAKULTA
ÚSTAV BIOCHEMIE

Testování efektu AI Foundational models na analýzu NGS dat DNA sekvenování

Diplomová práce

TOMÁŠ VAMBERSKÝ

Vedoucí práce: Mgr. Vojtěch Bystrý, Ph.D. Brno 2026

Bibliografický záznam

Autorka:	Bc. Tomáš Vamberský Přírodovědecká fakulta, Masarykova univerzita Ústav biochemie
Název práce:	Testování efektu AI Foundational models na analýzu NGS dat DNA sekvenování
Studijní program:	Biochemie
Studijní obor:	Bioinformatika
Vedoucí práce:	Mgr. Vojtěch Bystrý, Ph.D.
Akademický rok:	2025/2026
Počet stran:	?? + ??
Klíčová slova:	AI; Foundation model; Transcription factor

Bibliographic Entry

Author: Bc. Tomáš Vamberský
Faculty of Science, Masaryk University
Department of Biochemistry

Title of Thesis: Testing AI Foundational Models for Impact on NGS DNA Sequencing Data Analysis

Degree Programme: Biochemistry

Field of Study: Bioinformatics

Supervisor: Mgr. Vojtěch Bystrý, Ph.D.

Academic Year: 2025/2026

Number of Pages: ?? + ??

Keywords: AI; Foundation model; Transkripční faktor

Abstrakt

V této bakalářské/diplomové/rigorózní práci se věnujeme ...

Abstract

In this thesis we study ...

Místo tohoto listu vložte kopii oficiálního zadání práce bez podpisů.

Poděkování

Na tomto místě bych chtěl(-a) poděkovat ...

Prohlášení

Prohlašuji, že jsem svouji diplomovou práci vypracoval samostatně pod vedením vedoucího práce s využitím informačních zdrojů, které jsou v práci citovány. Prohlašuji, že jsem pro stylistické úpravy textu využil nástroje AI a to plně v souladu s principy akademické integrity.

Brno 13. Května 2026

.....
Tomáš Vamberský

Table of Contents

List of Figures	x
List of Tables	xi
List of Abbreviations	xii
Introduction	1
Chapter 1. Theoretical background	2
1.1 Transcription factor binding	2
1.1.1 Transcription factors and DNA binding	2
1.1.2 Sequence determinants of binding	2
1.1.3 The specificity paradox	3
1.2 Next-generation sequencing and ChIP-seq	4
1.3 Computational methods for motif analysis	4
1.3.1 PWMs	4
1.4 Foundation models for genomics	6
1.4.1 the architectures i guess	6
1.4.2 Evo2 and how they used it	6
Chapter 2. Aims of thesis	7
Chapter 3. Methods	8
3.1 Data	8
3.1.1 Transcription factors	8
3.1.2 ChIP-seq data sources	8
3.1.3 Cell-line coverage	9
3.1.4 Random-windows control	9
3.1.5 Reproducibility and seeding	10
3.2 Pipeline	10
3.2.1 Sequence extraction	10
3.2.2 Embedding extraction	11
3.2.3 Computational environment	11
Chapter 4. Druhá kapitola s matematikou $\int f(x) dx$ v názvu	13
4.1 Podkapitola	16

Závěr	17
Bibliography	22
Appendix	23

List of Figures

List of Tables

3.1 Chosen ChIP-seq experiments	9
---	---

List of abbreviations

- $\mathbb{C}$ množina všech komplexních čísel
- $\mathbb{R}$ množina všech reálných čísel
- $\mathbb{Z}$ množina všech celých čísel
- $\mathbb{N}$ množina všech přirozených čísel

Introduction

Chapter 1

Theoretical background

1.1 Transcription factor binding

1.1.1 Transcription factors and DNA binding

Transcription factors (TFs) are proteins that modulate transcription by binding DNA in a sequence-specific manner. They act through regulatory elements such as promoters, enhancers, and silencers, recruiting the transcription machinery or modulating its activity. Most TFs share a modular architecture: a DNA-binding domain (DBD) that contacts the target sequence, paired with one or more effector domains responsible for transactivation, repression, or dimerisation. Because the DBD determines binding specificity, TFs are classified by DBD family rather than by effector function. The human genome encodes approximately 1600 sequence-specific TFs distributed across roughly 75 DBD families (Lambert et al., 2018). The families represented among the TFs studied in this thesis include the C2H2 zinc fingers (the largest human family, exemplified by CTCF), the basic helix-loop-helix proteins (MYC, which binds as an obligate heterodimer with MAX), and the ETS domain family (SPI1).

Target sequences are typically short motifs of 6–20 bp, recognized with tolerance for sequence variation. Despite the considerable structural variety across DBD families, all sequence-specific TFs face the same fundamental problem: they must locate their short target sequence among a large number of similar but non-functional sites distributed across a genome of several gigabases.

1.1.2 Sequence determinants of binding

The specificity of protein–DNA recognition rests on two complementary mechanisms: direct chemical recognition of bases (base readout) and recognition of sequence-dependent DNA geometry (shape readout).

Base readout operates through direct contacts — hydrogen bonds and van der Waals interactions — between amino acid side chains and the Watson–Crick edges of bases, predominantly in the major groove. *The major groove presents a distinctive pattern of hydrogen bond donors and acceptors* that permits discrimination between all four base pairs, including the orientation of A:T and T:A (Luscombe et al., 2001; Seeman et al.,

1976). The C2H2 zinc finger illustrates the mechanism: each $\beta\beta\alpha$ module contacts approximately three base pairs through specificity residues on the recognition helix, and the contributions of successive modules combine to determine the affinity and specificity of the full array. Position weight matrices (PWMs) are the natural statistical encoding of base readout, treating each position within the motif as an independent contribution to binding affinity (§??).

Shape readout depends on sequence context beyond the contact position. The geometry at any given base pair — minor groove width, propeller twist, helical twist, roll, and slide — is influenced by neighbouring bases over several positions in either direction, and is read by protein contacts that respond to local shape and electrostatics rather than to specific bases. The canonical example is recognition of narrow minor grooves by arginine residues: runs of A:T base pairs (A-tracts) produce locally narrowed minor grooves with concentrated negative electrostatic potential, which arginine side chains insert into and stabilise without forming base-specific hydrogen bonds (Rohs et al., 2009).

The two mechanisms are genuinely complementary rather than redundant. Crystal structures of TF–DNA complexes routinely show major-groove base-specific contacts and minor-groove shape-based contacts within the same interface, and augmenting PWM-based models with predicted shape features systematically improves binding prediction across TF families (Mathelier et al., 2016). Their combined reach is nonetheless bounded by the binding footprint: base readout depends on direct contacts within the 6–20 bp footprint, and shape readout, while sensitive to flanking sequence, is mediated through contacts within and immediately adjacent to it (Yang et al., 2017).

1.1.3 The specificity paradox

The degeneracy of TF motifs means that a typical 10 bp motif at moderate information content matches millions of sites in a mammalian genome, while in vivo occupancy — assayed by ChIP-seq, the standard method described in §?? — typically identifies between a few thousand and a few tens of thousands of bound sites per TF per cell line. CTCF in K562 cells, for example, has approximately 52 000 IDR-thresholded ENCODE peaks, against approximately 870 000 matches to the canonical CTCF motif in the hg19 genome at standard FIMO thresholds (Dozmorov et al., 2022). This discrepancy between motif presence and actual occupancy has been termed the transcription factor specificity paradox; it was formalised for prokaryotic TFs as the futility theorem by Wunderlich and Mirny (2009), who showed that in a sufficiently large genome even highly informative motifs predict more matches than there are TF molecules in the cell, making purely sequence-based recognition statistically insufficient. In practice, the paradox is reflected in the weak correlation between intrinsic in vitro affinity (measured by HT-SELEX or protein-binding microarrays) and in vivo occupancy: high-affinity sites can remain unbound, while low-affinity sites are often occupied (Slattery et al., 2014).

Several mechanisms partially resolve the discrepancy, each a substantial field in its own right. Chromatin accessibility is regarded as the dominant factor: only a small fraction of the genome is accessible to DNA-binding proteins at any given time, the remainder occluded by nucleosomes or compacted into heterochromatin, and most in

vivo TF binding occurs within accessible regions (Thurman et al., 2012). Cooperative binding with cofactors restricts binding further: many TFs bind productively only as part of a complex — homodimers, heterodimers, or larger assemblies — so the combinatorics of cofactor co-occupancy limit where productive binding can occur (Sönmezer et al., 2021). Cellular context adds a third layer: TF abundance, post-translational modification, and subcellular localisation are all condition-dependent and regulate the effective concentration of binding-competent protein. **[TODO: yeah? where did i get this?]**

A fourth contribution, less studied than the preceding three, is sequence context beyond the motif itself. Dror et al. (2015) demonstrated, in a systematic analysis of 239 TFs from HT-SELEX and 56 from ChIP-seq data, that the sequence and DNA shape environment surrounding bound motifs differs significantly from that surrounding unbound occurrences of the same motif, with differences extending well beyond the core binding site and exhibiting TF-family-specific patterns. **[TODO: distinguish from shape readout somehow, read dror again]** Crucially, the signal was present in both the in vitro and in vivo datasets, ruling out the cellular environment as its sole source and establishing flanking sequence context as a genuine, sequence-encoded contributor to binding discrimination. Unlike chromatin accessibility and cooperative binding, this contribution is in principle directly accessible to any model that operates on raw sequence input.

The spatial scale at which sequence context influences binding spans several orders of magnitude: from the immediate flanks of the core motif (Dror et al., 2015), through the tens-to-hundreds of base pairs over which sequence composition modulates the nonspecific affinity experienced by a scanning TF (Afek a Lukatsky, 2013; Sela a Lukatsky, 2011), *out to the kilobase-scale compositional gradients recently characterised by Faltejsková et al. (2026)*. That last study, and its role as the biological motivation for the analyses in this thesis, is discussed in §?? following the description of the experimental data on which both studies are based. **[TODO: soften claim - descriptive? implied?, is this first mention? mention it's a preprint?]**

1.2 Next-generation sequencing and ChIP-seq

1.3 Computational methods for motif analysis

1.3.1 PWMs

[TODO: logo sekvence, vizuál pwm?]

The standard representation of a TF binding motif is the position weight matrix (PWM), constructed from an aligned set of known binding sequences. Counts of each base at each position give the position frequency matrix; dividing by the number of sequences gives the position probability matrix $f_{b,i}$. The PWM entries are the log-odds of these probabilities against a background distribution, $w_{b,i} = \log_2(f_{b,i}/p_b)$, where p_b is typically 0.25 for each base or the genome-wide composition. The conservation at each position is summarised by the information content $I_i = 2 - H_i$, where $H_i = -\sum_b f_{b,i} \log_2 f_{b,i}$ is the Shannon entropy of the base distribution at position i . Sequence

logos visualise the PWM by stacking letters at each position with heights proportional to $f_{b,i} \cdot I_i$, so that conserved positions appear as tall, near-pure stacks and degenerate positions as short, mixed ones. To identify candidate binding sites in a genome, the PWM is slid across the sequence and each L -bp window is scored by summing the matrix entries corresponding to its bases: $S = \sum_{i=1}^L w_{b_i,i}$. Standard scanning tools such as FIMO (Grant et al., 2011) convert this score into a p-value against the background distribution, and report windows below a chosen significance threshold as hits. The additive form of the scoring rule encodes a strong assumption: that each position contributes to binding affinity independently of the others. This approximation is known to be violated for many TFs — for example, where the identity of one position constrains which bases are tolerated at a neighbouring position [TODO: citaci] — and motivates the extensions discussed below.

Curated PWM models for known TFs are derived computationally from sets of enriched sequences, typically obtained from in vitro selection assays such as HT-SELEX or from high-confidence ChIP-seq peaks. The two most widely used discovery tools are MEME (Bailey et al., 2015), which identifies over-represented motifs by expectation maximisation, and HOMER (Heinz et al., 2010), which is optimised for ChIP-seq and identifies motifs by differential enrichment against a matched background. Both produce PWMs as their primary output. Curated databases — JASPAR (Ovek Baydar et al., 2026) for general-purpose use and HOCOMOCO (Vorontsov et al., 2024) for human and mouse TFs — collect such models across studies and provide reference resources for downstream applications; this thesis uses HOCOMOCO v12, matching the choice of (Faltejsková et al., 2026).

Several approaches relax the positional independence assumption of PWMs. Dinucleotide weight matrices extend the framework minimally, by scoring adjacent pairs of bases rather than single positions, and so capturing nearest-neighbour dependencies (Siebert a Söding, 2016); hidden Markov models go further, allowing variable-length and internally structured motifs through state transitions (Eddy, 2004) [TODO: terrible source and i dont have access, fix!]. Other extensions — Bayesian networks for arbitrary positional dependencies, and k-mer-based methods that learn from short subsequence frequencies directly — explore the same conceptual space. Benchmarking on in vitro binding data has shown that mononucleotide PWMs trained by the best methods perform comparably to more complex models for the majority of TFs, with measurable gains concentrated in a minority of cases (Weirauch et al., 2013). The fundamental constraint these extensions share is locality: regardless of how the independence assumption is relaxed, the model remains bounded to the motif region itself.

The PWM framework and its extensions are bounded by the binding footprint by design. The kilobase-scale gradients described in [TODO: inside reference, co s nimi?] §1.1.4, however, lie well outside that footprint, and have so far been characterised only with hand-engineered compositional features rather than learned directly from sequence. DNA language models — foundation models trained on raw genomic sequence over long contexts — operate at a scale where such gradients fall within their input window, and so could in principle encode them as part of their learned representation. The question this thesis addresses is whether a model of this kind, trained on raw sequence with no compositional features supplied, encodes the kilobase-scale signal that Faltejsková

et al. (2026) characterised by other means.

1.4 Foundation models for genomics

Foundation models are large neural networks trained on massive unlabelled datasets using self-supervised objectives, yielding general-purpose representations that downstream tasks can draw on (Bommasani et al., 2022). The two principal objectives are masked language modelling, in which the model recovers hidden tokens from their context, and autoregressive next-token prediction, in which the model predicts each token in sequence given everything to its left. This contrasts with supervised learning, which requires task-specific labels and therefore scales with the cost of annotation rather than with the availability of raw data. In genomics, this approach has produced models trained on raw nucleotide sequence at scale.

1.4.1 the architectures i guess

Genomic foundation models fall into two architectural families that have developed in parallel: those built on the transformer, and those built on alternatives to attention. The first genomic foundation models adapted BERT to nucleotide input: DNABERT (Ji et al., 2021) with masked language modelling on k-merised human sequence, and Nucleotide Transformer (Dalla-Torre et al., 2025) scaling this across multiple species and parameter counts. Both were limited in the length of input sequence they could process at once. In contrast, HyenaDNA (Nguyen et al., 2023) replaced attention with the Hyena operator, reaching context lengths in the hundreds of thousands of base pairs. Across both families, the design priority has been to extend the context window.

1.4.2 Evo2 and how they used it

Evo 2 (Brix et al., 2026) is a family of genomic foundation models developed by the Arc Institute, built on the StripedHyena 2 architecture. Attention scales quadratically in sequence length, which limits transformer-only models; StripedHyena 2 addresses this by combining attention with gated convolutional operators that scale linearly while preserving long-range dependencies. Evo 2's context window extends to 1 Mb at single-nucleotide resolution, comfortably hosting the kilobase-scale gradients of §1.1.4 within a single input. Evo 2 was pretrained autoregressively on OpenGenome2, a dataset of 8.8 trillion DNA tokens drawn from bacteria, archaea, eukarya, and bacteriophage. Pretraining proceeded in two phases: a short-context phase on roughly 8 kb windows enriched for genic regions, followed by a long-context phase extending to 1 Mb on whole-genome samples. Three capabilities follow: sequence scoring, which assigns a likelihood to an input under the model; generation, which samples new sequences; and embedding extraction, which retrieves representations from intermediate layers.

[TODO: simplified StripedHyena 2 diagram, or the one from the paper]

[TODO: what they did with TFs, what am i doing?]

Chapter 2

Aims of thesis

- 1.

Chapter 3

Methods

3.1 Data

3.1.1 Transcription factors

The descriptive analysis is performed on three human transcription factors selected to span distinct binding modes relevant to the questions posed in this thesis: CTCF, MYC, and SPI1. CTCF is an eleven-zinc-finger architectural factor with a long, well-characterised consensus motif and broad genomic occupancy, included as a strong-motif baseline against which embedding-based representations can be evaluated (Ong a Corces, 2014) MYC is a basic helix-loop-helix leucine-zipper factor that, in the analysis of Faltejsková et al. (2026), displays a pronounced directional dinucleotide asymmetry in the kilobase-scale flanks of its in vivo binding sites, making it a natural target for testing whether long-range sequence structure is reflected in Evo 2 representations (Lüscher a Vervoorts, 2012). SPI1 (PU.1) is an ETS-family lineage-specific factor that does not exhibit the directional flanking signature reported for MYC and is therefore included as a contrast factor for which the kilobase-scale signal is expected to be weaker or absent (Rothenberg et al., 2019).

3.1.2 ChIP-seq data sources

For each transcription factor, three ChIP-seq experiments are obtained from the ENCODE portal (Dunham et al., 2012), yielding nine BED files in total **[TODO: cite ENCODE portal / ENCODE3 if a more recent reference is preferred]**. Conservative IDR-thresholded narrowPeak files (Li et al., 2011) are used where available; otherwise, the IDR-thresholded peak file is used. Cell-line selection follows the inclusion criterion adopted by Faltejsková et al. (2026), which retains only cell lines appearing in at least three experiments across their panel. The accessions, transcription factors, and bio-samples comprising the embedded set are listed in Table ??, alongside the number of peaks supplied to the embedding pipeline. Sequence is extracted from the human reference assembly hg19 to maintain consistency with Faltejsková et al. (2026) and to permit direct comparison of per-bin readouts in the sanity check described in Section ??; this thesis does not consume any of the hg38-based annotation tracks (non-B DB, CpG

Tabulka 3.1: Chosen ChIP-seq experiments

TF	Cell line	ENCODE accession
MYC	GM12878	ENCFF765CKK
MYC	H1	ENCFF700CXD
MYC	MCF-7	ENCFF149BIS
CTCF	GM12878	ENCFF017XLW
CTCF	H1	ENCFF692RPA
CTCF	K562	ENCFF769AUF
SPI1	GM12878	ENCFF881ARS
SPI1	GM12878	ENCFF198SPL
SPI1	K562	ENCFF185DGL

islands, ENSEMBL regulatory features) used in the published study [TODO: cite hg19 / GRCh37 (UCSC or Genome Reference Consortium)]. BED files are downloaded programmatically from the ENCODE portal using the URL schema <https://www.encodeproject.org/f> [TODO: fix wrapping?]

3.1.3 Cell-line coverage

The intended design covers the cell lines GM12878, H1, and K562 across all three transcription factors, providing a balanced grid for within-TF and across-TF comparisons. The realised design departs from this target in one respect: the MYC K562 file (ENCFF043WTJ) was not embedded within the available compute window, and the MYC MCF-7 file (ENCFF149BIS) was substituted as the third MYC experiment. The resulting cell-line coverage is therefore CTCF on GM12878, H1, K562; MYC on GM12878, H1, MCF-7; and SPI1 on GM12878 (two replicates) and K562. The interpretive consequences of this imbalance for any TF–cell-line interaction analysis are addressed in the Results and Discussion chapters.

3.1.4 Random-windows control

A random-windows control is constructed to provide a non-peak-centred reference that shares the window geometry, sequence preprocessing, and embedding pipeline of the ChIP-seq data and against which TF-centred positional structure can be contrasted. Two thousand 10 kb windows are sampled from hg19 autosomes and chromosome X, with chromosome selection weighted by usable sequence length and chromosomes Y and M excluded. Candidate windows are subjected to rejection sampling against the union of MYC and CTCF peak sets and against previously accepted control windows, ensuring the control is disjoint from those peak sets and internally non-overlapping. SPI1 peaks are not included in the rejection set: the control was generated before SPI1 was added to the design and was not subsequently regenerated; given the small genomic footprint of the SPI1 peak set the residual chance of overlap is bounded but non-zero. A fixed random seed of 42 is used throughout.

3.1.5 Reproducibility and seeding

Stochastic elements of the pipeline are seeded explicitly to support reproducibility of intermediate quantities. A fixed random seed of 42 is applied at every sampling and stochastic-fitting site, including peak subsampling, random-windows generation, and UMAP fitting; this differs from the procedure of Faltejsková et al. (2026), in which peak subsampling is performed without a fixed seed. The full source code, configuration files, and environment-setup scripts used to produce the embedded set and the analyses reported in subsequent chapters are available at the project repository <https://github.com/TypeTwo30312/tv-thesis>. **[TODO: probably footnote the repo]**

3.2 Pipeline

This part describes the deterministic chain from ENCODE BED files to per-peak embedding tensors stored on disk. The three stages—sequence extraction, embedding extraction, and the computational environment in which they run—are each implemented as a separate Jupyter notebook in the project repository, and are reported here in the order in which they execute.

3.2.1 Sequence extraction

Window extraction reads an ENCODE narrowPeak file and outputs a per-peak record containing the peak coordinates and a fixed-length DNA sequence centred on the binding site. The 10 kb window length is chosen to match the ± 5000 bp range over which Faltejsková et al. (2026) report statistically significant compositional structure around in vivo binding sites. The peak midpoint is used as the binding-site proxy: for the ENCODE narrowPeak files inspected in this thesis, the reported summit coincides with the midpoint of the peak interval (CTCF GM12878 peaks are uniformly ~ 356 bp wide, MYC GM12878 peaks ~ 390 bp, MYC H1 ~ 264 bp), and the midpoint convention matches that used by Faltejsková et al. (2026). Peaks whose 10 kb window would extend past a chromosome boundary are dropped rather than padded, since padding with non-biological characters would introduce a fixed-position bias into any subsequent positional analysis. Non-ACGT characters in the extracted sequence are masked to N at the extraction stage; the analysis-stage filtering of windows containing fully-N 50 bp bins is described in Part 3 (Section ??). **[TODO: adjust ?? when Part 3 is drafted]** For each BED file, peaks are uniformly subsampled to a maximum of 10 000 (all peaks are retained if fewer are available), matching the MAX_SIZE_SAMPLED parameter used by Faltejsková et al. (2026) but with a fixed random seed of 42 throughout. Each output row carries the columns peak_id, chr, start, end, center, and sequence, and is written as a gzip-compressed CSV. The ACCESSION__TF__BIOSAMPLE naming convention used throughout the pipeline is preserved from the published source code released alongside Faltejsková et al. (2026), so that the per-bin feature-extraction scripts from that work remain directly applicable to the sanity check reported in Part 3 (Section ??). **[TODO: adjust ?? when Part 3 is drafted]**

3.2.2 Embedding extraction

Embedding extraction is performed with the Evo 2 7B checkpoint released by Brix et al. (2026); the architecture and pretraining objective are described in the background chapter (Section ??). **[TODO: adjust ?? to the actual background-chapter label]** The 7B family is used because it runs natively in bfloat16 on Ampere-class hardware and does not require Transformer Engine or FP8 support, which are needed for the 20B and 40B variants and are unavailable on the GPUs accessible through the CERIT-SC cluster used in this work. **[TODO: not true!! fix, reasoning]** Within the 7B family, the 1 M-context variant (evo2_7b) is selected over the 8 kb-context base variant (evo2_7b_base) because the chosen 10 kb window exceeds the latter's context length. A single intermediate-layer embedding is extracted per token from the layer blocks.28.mlp.13; intermediate layers are recommended over the final layer for representation learning because the final layer is specialised to the next-token prediction objective, and blocks.28.mlp.13 is the layer used by the exon-classifier example in the official Evo 2 repository (Brix et al., 2026). **[TODO: either run a comparison or suggest it in discussion]** Each 10 kb sequence is tokenised under the Evo 2 byte-level scheme into a (1, 10000) integer tensor (one token per nucleotide), passed through the model in a single forward pass on one GPU, and the resulting (1, 10000, 4096) bfloat16 activation tensor is reshaped into 200 non-overlapping 50 bp bins and averaged within each bin to yield a (200, 4096) representation per peak, which is cast to fp16 before writing. The 50 bp bin width matches the k -mer resolution adopted by Faltejsková et al. (2026) so that per-bin readouts are directly comparable; non-overlapping bins are used in place of their 25 bp stride because each bin already integrates 50 input positions of the same forward pass and stride overlap would not contribute additional information. Unlike the strictly local hand-engineered features used by Faltejsková et al. (2026), each binned embedding has the full 10 kb window in its causal receptive field via the autoregressive forward pass, and so reflects model-internal aggregation of context up to that point rather than a fixed-window feature of the sequence itself. A streaming-write strategy based on `numpy.memmap` is used for the per-file accumulator together with a checkpoint every 500 peaks, because retaining all per-peak tensors in memory before writing exceeds the per-pod RAM allocation on the cluster for the larger BED files. The output of this stage is, per BED file, a single .npz archive containing the $(n_{\text{peaks}}, 200, 4096)$ fp16 array together with a .parquet sidecar carrying the peak_id column and BED-derived metadata in the same row order. Embedding extraction was performed on a single NVIDIA A40 GPU (46 GB VRAM) with peak allocation of 19.17 GB at batch size 1, at an observed throughput of approximately 0.5 peaks per second; the embedded set comprises the 9 ChIP-seq files of Part 1 plus the random-windows control of Section 3.1.4, totalling 96.9 GB of compressed embeddings on disk.

3.2.3 Computational environment

The embedding stage runs on the CERIT-SC JupyterHub platform under the NVIDIA PyTorch 2.5.0 container image, which provides a Python 3.10, CUDA 12.6, and Ubuntu 22.04 base; a project-specific Python 3.12 virtual environment is layered on top, installed via uv into the persistent home directory and pinning torch 2.7.1+cu128, flash-attn 2.8.0, pos

and `evo2`. A separate CPU-only environment is used for the analyses reported in Part 3, isolating the heavy GPU stack from sessions that do not require it and pinning `numpy`, `polars`, `scikit-learn`, and `umap-learn` alongside the standard scientific Python packages. Both environments are reproduced by setup scripts (`setup_evo2_env.sh`, `setup_exploration` **[TODO: nope! single evo2 environment, no setup scripts]** maintained alongside the source code, and the full code base, configuration files, and notebooks are available at the project repository¹.

¹<https://github.com/TypeTwo30312/tv-thesis>

Druhá kapitola s matematikou $\int f(x) dx$ v názvu

$$\int f(x) \mathrm{d}x \quad (\text{rovnice})$$
$$\iint f(x) \mathrm{d}x \quad (4.1)$$

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

- 13 -

nunc eu urna hendrerit faucibus. Aliquam auctor, pede consequat laoreet varius, eros tellus scelerisque quam, pellentesque hendrerit ipsum dolor sed augue. Nulla nec lacus.

Suspendisse vitae elit. Aliquam arcu neque, ornare in, ullamcorper quis, commodo eu, libero. Fusce sagittis erat at erat tristique mollis. Maecenas sapien libero, molestie et, lobortis in, sodales eget, dui. Morbi ultrices rutrum lorem. Nam elementum ullamcorper leo. Morbi dui. Aliquam sagittis. Nunc placerat. Pellentesque tristique sodales est. Maecenas imperdiet lacinia velit. Cras non urna. Morbi eros pede, suscipit ac, varius vel, egestas non, eros. Praesent malesuada, diam id pretium elementum, eros sem dictum tortor, vel consectetur odio sem sed wisi.

Sed feugiat. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Ut pellentesque augue sed urna. Vestibulum diam eros, fringilla et, consectetur eu, nonummy id, sapien. Nullam at lectus. In sagittis ultrices mauris. Curabitur malesuada erat sit amet massa. Fusce blandit. Aliquam erat volutpat. Aliquam euismod. Aenean vel lectus. Nunc imperdiet justo nec dolor.

Etiam euismod. Fusce facilisis lacinia dui. Suspendisse potenti. In mi erat, cursus id, nonummy sed, ullamcorper eget, sapien. Praesent pretium, magna in eleifend egestas, pede pede pretium lorem, quis consectetur tortor sapien facilisis magna. Mauris quis magna varius nulla scelerisque imperdiet. Aliquam non quam. Aliquam porttitor quam a lacus. Praesent vel arcu ut tortor cursus volutpat. In vitae pede quis diam bibendum placerat. Fusce elementum convallis neque. Sed dolor orci, scelerisque ac, dapibus nec, ultricies ut, mi. Duis nec dui quis leo sagittis commodo.

Aliquam lectus. Vivamus leo. Quisque ornare tellus ullamcorper nulla. Mauris porttitor pharetra tortor. Sed fringilla justo sed mauris. Mauris tellus. Sed non leo. Nullam elementum, magna in cursus sodales, augue est scelerisque sapien, venenatis congue nulla arcu et pede. Ut suscipit enim vel sapien. Donec congue. Maecenas urna mi, suscipit in, placerat ut, vestibulum ut, massa. Fusce ultrices nulla et nisl.

Etiam ac leo a risus tristique nonummy. Donec dignissim tincidunt nulla. Vestibulum rhoncus molestie odio. Sed lobortis, justo et pretium lobortis, mauris turpis condimentum augue, nec ultricies nibh arcu pretium enim. Nunc purus neque, placerat id, imperdiet sed, pellentesque nec, nisl. Vestibulum imperdiet neque non sem accumsan laoreet. In hac habitasse platea dictumst. Etiam condimentum facilisis libero. Suspendisse in elit quis nisl aliquam dapibus. Pellentesque auctor sapien. Sed egestas sapien nec lectus. Pellentesque vel dui vel neque bibendum viverra. Aliquam porttitor nisl nec pede. Proin mattis libero vel turpis. Donec rutrum mauris et libero. Proin euismod porta felis. Nam lobortis, metus quis elementum commodo, nunc lectus elementum mauris, eget vulputate ligula tellus eu neque. Vivamus eu dolor. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Aenean nonummy turpis id odio. Integer euismod imperdiet turpis. Ut nec leo nec diam imperdiet lacinia. Etiam eget lacus eget mi ultricies posuere. In placerat tristique tortor. Sed porta vestibulum metus. Nulla iaculis sollicitudin pede. Fusce luctus tellus in dolor. Curabitur auctor velit a sem. Morbi sapien. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Donec adipiscing urna vehicula nunc. Sed ornare leo in leo. In rhoncus leo ut dui. Aenean dolor quam, volutpat nec, fringilla id, consectetur vel, pede.

Nulla malesuada risus ut urna. Aenean pretium velit sit amet metus. Duis iaculis. In

hac habitasse platea dictumst. Nullam molestie turpis eget nisl. Duis a massa id pede dapibus ultricies. Sed eu leo. In at mauris sit amet tortor bibendum varius. Phasellus justo risus, posuere in, sagittis ac, varius vel, tortor. Quisque id enim. Phasellus consequat, libero pretium nonummy fringilla, tortor lacus vestibulum nunc, ut rhoncus ligula neque id justo. Nullam accumsan euismod nunc. Proin vitae ipsum ac metus dictum tempus. Nam ut wisi. Quisque tortor felis, interdum ac, sodales a, semper a, sem. Curabitur in velit sit amet dui tristique sodales. Vivamus mauris pede, lacinia eget, pellentesque quis, scelerisque eu, est. Aliquam risus. Quisque bibendum pede eu dolor.

Donec tempus neque vitae est. Aenean egestas odio sed risus ullamcorper ullamcorper. Sed in nulla a tortor tincidunt egestas. Nam sapien tortor, elementum sit amet, aliquam in, porttitor faucibus, enim. Nullam congue suscipit nibh. Quisque convallis. Praesent arcu nibh, vehicula eget, accumsan eu, tincidunt a, nibh. Suspendisse vulputate, tortor quis adipiscing viverra, lacus nibh dignissim tellus, eu suscipit risus ante fringilla diam. Quisque a libero vel pede imperdiet aliquet. Pellentesque nunc nibh, eleifend a, consequat consequat, hendrerit nec, diam. Sed urna. Maecenas laoreet eleifend neque. Vivamus purus odio, eleifend non, iaculis a, ultrices sit amet, urna. Mauris faucibus odio vitae risus. In nisl. Praesent purus. Integer iaculis, sem eu egestas lacinia, lacus pede scelerisque augue, in ullamcorper dolor eros ac lacus. Nunc in libero.

Fusce suscipit cursus sem. Vivamus risus mi, egestas ac, imperdiet varius, faucibus quis, leo. Aenean tincidunt. Donec suscipit. Cras id justo quis nibh scelerisque dignissim. Aliquam sagittis elementum dolor. Aenean consectetur justo in pede. Curabitur ullamcorper ligula nec orci. Aliquam purus turpis, aliquam id, ornare vitae, porttitor non, wisi. Maecenas luctus porta lorem. Donec vitae ligula eu ante pretium varius. Proin tortor metus, convallis et, hendrerit non, scelerisque in, urna. Cras quis libero eu ligula bibendum tempor. Vivamus tellus quam, malesuada eu, tempus sed, tempor sed, velit. Donec lacinia auctor libero.

Praesent sed neque id pede mollis rutrum. Vestibulum iaculis risus. Pellentesque lacus. Ut quis nunc sed odio malesuada egestas. Duis a magna sit amet ligula tristique pretium. Ut pharetra. Vestibulum imperdiet magna nec wisi. Mauris convallis. Sed accumsan sollicitudin massa. Sed id enim. Nunc pede enim, lacinia ut, pulvinar quis, suscipit semper, elit. Cras accumsan erat vitae enim. Cras sollicitudin. Vestibulum rutrum blandit massa.

Sed gravida lectus ut purus. Morbi laoreet magna. Pellentesque eu wisi. Proin turpis. Integer sollicitudin augue nec dui. Fusce lectus. Vivamus faucibus nulla nec lacus. Integer diam. Pellentesque sodales, enim feugiat cursus volutpat, sem mauris dignissim mauris, quis consequat sem est fermentum ligula. Nullam justo lectus, condimentum sit amet, posuere a, fringilla mollis, felis. Morbi nulla nibh, pellentesque at, nonummy eu, sollicitudin nec, ipsum. Cras neque. Nunc augue. Nullam vitae quam id quam pulvinar blandit. Nunc sit amet orci. Aliquam erat elit, pharetra nec, aliquet a, gravida in, mi. Quisque urna enim, viverra quis, suscipit quis, tincidunt ut, sapien. Cras placerat consequat sem. Curabitur ac diam. Curabitur diam tortor, mollis et, viverra ac, tempus vel, metus.

Curabitur ac lorem. Vivamus non justo in dui mattis posuere. Etiam accumsan ligula id pede. Maecenas tincidunt diam nec velit. Praesent convallis sapien ac est. Aliquam ullamcorper euismod nulla. Integer mollis enim vel tortor. Nulla sodales placerat nunc.

Sed tempus rutrum wisi. Duis accumsan gravida purus. Nunc nunc. Etiam facilisis dui eu sem. Vestibulum semper. Praesent eu eros. Vestibulum tellus nisl, dapibus id, vestibulum sit amet, placerat ac, mauris. Maecenas et elit ut erat placerat dictum. Nam feugiat, turpis et sodales volutpat, wisi quam rhoncus neque, vitae aliquam ipsum sapien vel enim. Maecenas suscipit cursus mi.

4.1 Podkapitola

Quisque facilisis auctor sapien. Pellentesque gravida hendrerit lectus. Mauris rutrum sodales sapien. Fusce hendrerit sem vel lorem. Integer pellentesque massa vel augue. Integer elit tortor, feugiat quis, sagittis et, ornare non, lacus. Vestibulum posuere pellentesque eros. Quisque venenatis ipsum dictum nulla. Aliquam quis quam non metus eleifend interdum. Nam eget sapien ac mauris malesuada adipiscing. Etiam eleifend neque sed quam. Nulla facilisi. Proin a ligula. Sed id dui eu nibh egestas tincidunt. Suspendisse arcu.

Maecenas dui. Aliquam volutpat auctor lorem. Cras placerat est vitae lectus. Curabitur massa lectus, rutrum euismod, dignissim ut, dapibus a, odio. Ut eros erat, vulputate ut, interdum non, porta eu, erat. Cras fermentum, felis in porta congue, velit leo facilisis odio, vitae consectetur lorem quam vitae orci. Sed ultrices, pede eu placerat auctor, ante ligula rutrum tellus, vel posuere nibh lacus nec nibh. Maecenas laoreet dolor at enim. Donec molestie dolor nec metus. Vestibulum libero. Sed quis erat. Sed tristique. Duis pede leo, fermentum quis, consectetur eget, vulputate sit amet, erat.

Donec vitae velit. Suspendisse porta fermentum mauris. Ut vel nunc non mauris pharetra varius. Duis consequat libero quis urna. Maecenas at ante. Vivamus varius, wisi sed egestas tristique, odio wisi luctus nulla, lobortis dictum dolor ligula in lacus. Vivamus aliquam, urna sed interdum porttitor, metus orci interdum odio, sit amet euismod lectus felis et leo. Praesent ac wisi. Nam suscipit vestibulum sem. Praesent eu ipsum vitae pede cursus venenatis. Duis sed odio. Vestibulum eleifend. Nulla ut massa. Proin rutrum mattis sapien. Curabitur dictum gravida ante.

Phasellus placerat vulputate quam. Maecenas at tellus. Pellentesque neque diam, dignissim ac, venenatis vitae, consequat ut, lacus. Nam nibh. Vestibulum fringilla arcu mollis arcu. Sed et turpis. Donec sem tellus, volutpat et, varius eu, commodo sed, lectus. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Quisque enim arcu, suscipit nec, tempus at, imperdiet vel, metus. Morbi volutpat purus at erat. Donec dignissim, sem id semper tempus, nibh massa eleifend turpis, sed pellentesque wisi purus sed libero. Nullam lobortis tortor vel risus. Pellentesque consequat nulla eu tellus. Donec velit. Aliquam fermentum, wisi ac rhoncus iaculis, tellus nunc malesuada orci, quis volutpat dui magna id mi. Nunc vel ante. Duis vitae lacus. Cras nec ipsum.

Závěr

[illegible][illegible]

Quisque facilisis auctor sapien. Pellentesque gravida hendrerit lectus. Mauris rutrum sodales sapien. Fusce hendrerit sem vel lorem. Integer pellentesque massa vel augue. Integer elit tortor, feugiat quis, sagittis et, ornare non, lacus. Vestibulum posuere pellentesque eros. Quisque venenatis ipsum dictum nulla. Aliquam quis quam non metus eleifend interdum. Nam eget sapien ac mauris malesuada adipiscing. Etiam eleifend neque sed quam. Nulla facilisi. Proin a ligula. Sed id dui eu nibh egetas tincidunt. Suspendisse arcu.

Maecenas dui. Aliquam volutpat auctor lorem. Cras placerat est vitae lectus. Curabitur massa lectus, rutrum euismod, dignissim ut, dapibus a, odio. Ut eros erat, vulputate ut, interdum non, porta eu, erat. Cras fermentum, felis in porta congue, velit leo facilisis odio, vitae consectetur lorem quam vitae orci. Sed ultrices, pede eu placerat auctor, ante ligula rutrum tellus, vel posuere nibh lacus nec nibh. Maecenas laoreet dolor at enim. Donec molestie dolor nec metus. Vestibulum libero. Sed quis erat. Sed tristique. Duis pede leo, fermentum quis, consectetur eget, vulputate sit amet, erat.

Donec vitae velit. Suspendisse porta fermentum mauris. Ut vel nunc non mauris pharetra varius. Duis consequat libero quis urna. Maecenas at ante. Vivamus varius, wisi sed egestas tristique, odio wisi luctus nulla, lobortis dictum dolor ligula in lacus. Vivamus aliquam, urna sed interdum porttitor, metus orci interdum odio, sit amet euismod lectus felis et leo. Praesent ac wisi. Nam suscipit vestibulum sem. Praesent eu ipsum vitae pede cursus venenatis. Duis sed odio. Vestibulum eleifend. Nulla ut massa. Proin rutrum mattis sapien. Curabitur dictum gravida ante.

Phasellus placerat vulputate quam. Maecenas at tellus. Pellentesque neque diam, dignissim ac, venenatis vitae, consequat ut, lacus. Nam nibh. Vestibulum fringilla arcu mollis arcu. Sed et turpis. Donec sem tellus, volutpat et, varius eu, commodo sed, lectus. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Quisque enim arcu, suscipit

nec, tempus at, imperdiet vel, metus. Morbi volutpat purus at erat. Donec dignissim, sem id semper tempus, nibh massa eleifend turpis, sed pellentesque wisi purus sed libero. Nullam lobortis tortor vel risus. Pellentesque consequat nulla eu tellus. Donec velit. Aliquam fermentum, wisi ac rhoncus iaculis, tellus nunc malesuada orci, quis volutpat dui magna id mi. Nunc vel ante. Duis vitae lacus. Cras nec ipsum.

Morbi nunc. Aliquam consectetur varius nulla. Phasellus eros. Cras dapibus porttitor risus. Maecenas ultrices mi sed diam. Praesent gravida velit at elit vehicula porttitor. Phasellus nisl mi, sagittis ac, pulvinar id, gravida sit amet, erat. Vestibulum est. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Curabitur id sem elementum leo rutrum hendrerit. Ut at mi. Donec tincidunt faucibus massa. Sed turpis quam, sollicitudin a, hendrerit eget, pretium ut, nisl. Duis hendrerit ligula. Nunc pulvinar congue urna.

Nunc velit. Nullam elit sapien, eleifend eu, commodo nec, semper sit amet, elit. Nulla lectus risus, condimentum ut, laoreet eget, viverra nec, odio. Proin lobortis. Curabitur dictum arcu vel wisi. Cras id nulla venenatis tortor congue ultrices. Pellentesque eget pede. Sed eleifend sagittis elit. Nam sed tellus sit amet lectus ullamcorper tristique. Mauris enim sem, tristique eu, accumsan at, scelerisque vulputate, neque. Quisque lacus. Donec et ipsum sit amet elit nonummy aliquet. Sed viverra nisl at sem. Nam diam. Mauris ut dolor. Curabitur ornare tortor cursus velit.

Morbi tincidunt posuere arcu. Cras venenatis est vitae dolor. Vivamus scelerisque semper mi. Donec ipsum arcu, consequat scelerisque, viverra id, dictum at, metus. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut pede sem, tempus ut, porttitor bibendum, molestie eu, elit. Suspendisse potenti. Sed id lectus sit amet purus faucibus vehicula. Praesent sed sem non dui pharetra interdum. Nam viverra ultrices magna.

Aenean laoreet aliquam orci. Nunc interdum elementum urna. Quisque erat. Nullam tempor neque. Maecenas velit nibh, scelerisque a, consequat ut, viverra in, enim. Duis magna. Donec odio neque, tristique et, tincidunt eu, rhoncus ac, nunc. Mauris malesuada malesuada elit. Etiam lacus mauris, pretium vel, blandit in, ultricies id, libero. Phasellus bibendum erat ut diam. In congue imperdiet lectus.

Aenean scelerisque. Fusce pretium porttitor lorem. In hac habitasse platea dictumst. Nulla sit amet nisl at sapien egestas pretium. Nunc non tellus. Vivamus aliquet. Nam adipiscing euismod dolor. Aliquam erat volutpat. Nulla ut ipsum. Quisque tincidunt auctor augue. Nunc imperdiet ipsum eget elit. Aliquam quam leo, consectetur non, ornare sit amet, tristique quis, felis. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Pellentesque interdum quam sit amet mi. Pellentesque mauris dui, dictum a, adipiscing ac, fermentum sit amet, lorem.

Ut quis wisi. Praesent quis massa. Vivamus egestas risus eget lacus. Nunc tincidunt, risus quis bibendum facilisis, lorem purus rutrum neque, nec porta tortor urna quis orci. Aenean aliquet, libero semper volutpat luctus, pede erat lacinia augue, quis rutrum sem ipsum sit amet pede. Vestibulum aliquet, nibh sed iaculis sagittis, odio dolor blandit augue, eget mollis urna tellus id tellus. Aenean aliquet aliquam nunc. Nulla ultricies justo eget orci. Phasellus tristique fermentum leo. Sed massa metus, sagittis ut, semper ut, pharetra vel, erat. Aliquam quam turpis, egestas vel, elementum in, egestas sit amet, lorem. Duis convallis, wisi sit amet mollis molestie, libero mauris porta dui, vitae aliquam arcu turpis ac sem. Aliquam aliquet dapibus metus.

Vivamus commodo eros eleifend dui. Vestibulum in leo eu erat tristique mattis. Cras

at elit. Cras pellentesque. Nullam id lacus sit amet libero aliquet hendrerit. Proin placerat, mi non elementum laoreet, eros elit tincidunt magna, a rhoncus sem arcu id odio. Nulla eget leo a leo egestas facilisis. Curabitur quis velit. Phasellus aliquam, tortor nec ornare rhoncus, purus urna posuere velit, et commodo risus tellus quis tellus. Vivamus leo turpis, tempus sit amet, tristique vitae, laoreet quis, odio. Proin scelerisque bibendum ipsum. Etiam nisl. Praesent vel dolor. Pellentesque vel magna. Curabitur urna. Vivamus congue urna in velit. Etiam ullamcorper elementum dui. Praesent non urna. Sed placerat quam non mi. Pellentesque diam magna, ultricies eget, ultrices placerat, adipiscing rutrum, sem.

Morbi sem. Nulla facilisi. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Nulla facilisi. Morbi sagittis ultrices libero. Praesent eu ligula sed sapien auctor sagittis. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Donec vel nunc. Nunc fermentum, lacus id aliquam porta, dui tortor euismod eros, vel molestie ipsum purus eu lacus. Vivamus pede arcu, euismod ac, tempus id, pretium et, lacus. Curabitur sodales dapibus urna. Nunc eu sapien. Donec eget nunc a pede dictum pretium. Proin mauris. Vivamus luctus libero vel nibh.

Fusce tristique risus id wisi. Integer molestie massa id sem. Vestibulum vel dolor. Pellentesque vel urna vel risus ultricies elementum. Quisque sapien urna, blandit nec, iaculis ac, viverra in, odio. In hac habitasse platea dictumst. Morbi neque lacus, convallis vitae, commodo ac, fermentum eu, velit. Sed in orci. In fringilla turpis non arcu. Donec in ante. Phasellus tempor feugiat velit. Aenean varius massa non turpis. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae;

Aliquam tortor. Morbi ipsum massa, imperdiet non, consectetur vel, feugiat vel, lorem. Quisque eget lorem nec elit malesuada vestibulum. Quisque sollicitudin ipsum vel sem. Nulla enim. Proin nonummy felis vitae felis. Nullam pellentesque. Duis rutrum feugiat felis. Mauris vel pede sed libero tincidunt mollis. Phasellus sed urna rhoncus diam euismod bibendum. Phasellus sed nisl. Integer condimentum justo id orci iaculis varius. Quisque et lacus. Phasellus elementum, justo at dignissim auctor, wisi odio lobortis arcu, sed sollicitudin felis felis eu neque. Praesent at lacus.

Vivamus sit amet pede. Duis interdum, nunc eget rutrum dignissim, nisl diam luctus leo, et tincidunt velit nisl id tellus. In lorem tellus, aliquet vitae, porta in, aliquet sed, lectus. Phasellus sodales. Ut varius scelerisque erat. In vel nibh eu eros imperdiet rutrum. Donec ac odio nec neque vulputate suscipit. Nam nec magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Nullam porta, odio et sagittis iaculis, wisi neque fringilla sapien, vel commodo lorem lorem id elit. Ut sem lectus, scelerisque eget, placerat et, tincidunt scelerisque, ligula. Pellentesque non orci.

Etiam vel ipsum. Morbi facilisis vestibulum nisl. Praesent cursus laoreet felis. Integer adipiscing pretium orci. Nulla facilisi. Quisque posuere bibendum purus. Nulla quam mauris, cursus eget, convallis ac, molestie non, enim. Aliquam congue. Quisque sagittis nonummy sapien. Proin molestie sem vitae urna. Maecenas lorem. Vivamus viverra consequat enim.

Bibliography

- [1] Ariel Afek and David B. Lukatsky. *Genome-Wide Organization of Eukaryotic Pre-Initiation Complex Is Influenced by Nonconsensus Protein-DNA Binding*. <https://arxiv.org/abs/1303.0103>. Mar. 2013. DOI: [10.1016/j.bpj.2013.01.038](https://doi.org/10.1016/j.bpj.2013.01.038). (Visited on 05/04/2026).
- [2] Timothy L. Bailey et al. „The MEME Suite“. In: *Nucleic Acids Research* 43.W1 (July 2015), W39–W49. ISSN: 0305-1048. DOI: [10.1093/nar/gkv416](https://doi.org/10.1093/nar/gkv416). (Visited on 05/04/2026).
- [3] Rishi Bommasani et al. *On the Opportunities and Risks of Foundation Models*. July 2022. DOI: [10.48550/arXiv.2108.07258](https://doi.org/10.48550/arXiv.2108.07258). arXiv: [2108.07258 \[cs\]](https://arxiv.org/abs/2108.07258). (Visited on 05/04/2026).
- [4] Garyk Brixi et al. „Genome Modelling and Design across All Domains of Life with Evo 2“. In: *Nature* 652.8112 (Apr. 2026), pp. 1349–1361. ISSN: 1476-4687. DOI: [10.1038/s41586-026-10176-5](https://doi.org/10.1038/s41586-026-10176-5). (Visited on 05/05/2026).
- [5] Hugo Dalla-Torre et al. „Nucleotide Transformer: Building and Evaluating Robust Foundation Models for Human Genomics“. In: *Nature Methods* 22.2 (Feb. 2025), pp. 287–297. ISSN: 1548-7105. DOI: [10.1038/s41592-024-02523-z](https://doi.org/10.1038/s41592-024-02523-z). (Visited on 03/17/2025).
- [6] Mikhail G Dozmorov et al. „CTCF: An R/Bioconductor Data Package of Human and Mouse CTCF Binding Sites“. In: *Bioinformatics Advances* 2.1 (Dec. 2022), vbac097. ISSN: 2635-0041. DOI: [10.1093/bioadv/vbac097](https://doi.org/10.1093/bioadv/vbac097). (Visited on 04/30/2026). ■
- [7] Iris Dror et al. „A Widespread Role of the Motif Environment in Transcription Factor Binding across Diverse Protein Families“. In: *Genome Research* 25.9 (July 2015), pp. 1268–1280. ISSN: 1088-9051, 1549-5469. DOI: [10.1101/gr.184671.114](https://doi.org/10.1101/gr.184671.114). (Visited on 04/14/2026).
- [8] Ian Dunham et al. „An Integrated Encyclopedia of DNA Elements in the Human Genome“. In: *Nature* 489.7414 (Sept. 2012), pp. 57–74. ISSN: 1476-4687. DOI: [10.1038/nature11247](https://doi.org/10.1038/nature11247). (Visited on 05/04/2026).
- [9] Sean R. Eddy. „What Is a Hidden Markov Model?“ In: *Nature Biotechnology* 22.10 (Oct. 2004), pp. 1315–1316. ISSN: 1546-1696. DOI: [10.1038/nbt1004-1315](https://doi.org/10.1038/nbt1004-1315). (Visited on 05/04/2026).
- [10] Kateřina Faltejsová, Josef Šulc, and Jiří Vondrášek. *Non-Consensus Flanking Sequence of Hundreds of Base Pairs around in Vivo Binding Sites: Statistical Beacons for Transcription Factor Scanning*. Mar. 2026. DOI: [10.1101/2025.05.28.656598](https://doi.org/10.1101/2025.05.28.656598). (Visited on 03/14/2026).

- [11] Charles E. Grant, Timothy L. Bailey, and William Stafford Noble. „FIMO: Scanning for Occurrences of a given Motif“. In: *Bioinformatics* 27.7 (Apr. 2011), pp. 1017–1018. ISSN: 1367-4803. DOI: [10.1093/bioinformatics/btr064](https://doi.org/10.1093/bioinformatics/btr064). (Visited on 05/04/2026).
- [12] Sven Heinz et al. „Simple Combinations of Lineage-Determining Transcription Factors Prime *Cis*-Regulatory Elements Required for Macrophage and B Cell Identities“. In: *Molecular Cell* 38.4 (May 2010), pp. 576–589. ISSN: 1097-2765. DOI: [10.1016/j.molcel.2010.05.004](https://doi.org/10.1016/j.molcel.2010.05.004). (Visited on 05/04/2026).
- [13] Yanrong Ji et al. „DNABERT: Pre-Trained Bidirectional Encoder Representations from Transformers Model for DNA-language in Genome“. In: *Bioinformatics* 37.15 (Aug. 2021), pp. 2112–2120. ISSN: 1367-4803. DOI: [10.1093/bioinformatics/btab083](https://doi.org/10.1093/bioinformatics/btab083). (Visited on 05/04/2026).
- [14] Samuel A. Lambert et al. „The Human Transcription Factors“. In: *Cell* 172.4 (Feb. 2018), pp. 650–665. ISSN: 0092-8674. DOI: [10.1016/j.cell.2018.01.029](https://doi.org/10.1016/j.cell.2018.01.029). (Visited on 04/17/2026).
- [15] Qunhua Li et al. „Measuring Reproducibility of High-Throughput Experiments“. In: *The Annals of Applied Statistics* 5.3 (Sept. 2011), pp. 1752–1779. ISSN: 1932-6157, 1941-7330. DOI: [10.1214/11-AOAS466](https://doi.org/10.1214/11-AOAS466). (Visited on 05/04/2026).
- [16] N. M. Luscombe, R. A. Laskowski, and J. M. Thornton. „Amino Acid-Base Interactions: A Three-Dimensional Analysis of Protein-DNA Interactions at an Atomic Level“. In: *Nucleic Acids Research* 29.13 (July 2001), pp. 2860–2874. ISSN: 1362-4962. DOI: [10.1093/nar/29.13.2860](https://doi.org/10.1093/nar/29.13.2860).
- [17] Bernhard Lüscher and Jörg Vervoorts. „Regulation of Gene Transcription by the Oncoprotein MYC“. In: *Gene* 494.2 (Feb. 2012), pp. 145–160. ISSN: 0378-1119. DOI: [10.1016/j.gene.2011.12.027](https://doi.org/10.1016/j.gene.2011.12.027). (Visited on 05/07/2026).
- [18] Anthony Mathelier et al. „DNA Shape Features Improve Transcription Factor Binding Site Predictions In Vivo“. In: *Cell Systems* 3.3 (Sept. 2016), pp. 278–286.e4. ISSN: 2405-4712. DOI: [10.1016/j.cels.2016.07.001](https://doi.org/10.1016/j.cels.2016.07.001). (Visited on 04/30/2026).
- [19] Eric Nguyen et al. *HyenaDNA: Long-Range Genomic Sequence Modeling at Single Nucleotide Resolution*. Nov. 2023. DOI: [10.48550/arXiv.2306.15794](https://doi.org/10.48550/arXiv.2306.15794). arXiv: [2306.15794 \[cs\]](https://arxiv.org/abs/2306.15794). (Visited on 05/04/2026).
- [20] Chin-Tong Ong and Victor G. Corces. „CTCF: An Architectural Protein Bridging Genome Topology and Function“. In: *Nature reviews. Genetics* 15.4 (Apr. 2014), pp. 234–246. ISSN: 1471-0056. DOI: [10.1038/nrg3663](https://doi.org/10.1038/nrg3663). (Visited on 05/07/2026).
- [21] Damla Ovek Baydar et al. „JASPAR 2026: Expansion of Transcription Factor Binding Profiles and Integration of Deep Learning Models“. In: *Nucleic Acids Research* 54.D1 (Jan. 2026), pp. D184–D193. ISSN: 1362-4962. DOI: [10.1093/nar/gkaf1209](https://doi.org/10.1093/nar/gkaf1209).
- [22] Remo Rohs et al. „The Role of DNA Shape in Protein-DNA Recognition“. In: *Nature* 461.7268 (Oct. 2009), pp. 1248–1253. ISSN: 0028-0836. DOI: [10.1038/nature08473](https://doi.org/10.1038/nature08473). (Visited on 04/30/2026).

- [23] Ellen V. Rothenberg, Hiroyuki Hosokawa, and Jonas Ungerbäck. „Mechanisms of Action of Hematopoietic Transcription Factor PU.1 in Initiation of T-Cell Development“. In: *Frontiers in Immunology* 10 (Feb. 2019). ISSN: 1664-3224. DOI: [10.3389/fimmu.2019.00228](https://doi.org/10.3389/fimmu.2019.00228). (Visited on 05/07/2026).
- [24] N C Seeman, J M Rosenberg, and A Rich. „Sequence-Specific Recognition of Double Helical Nucleic Acids by Proteins.“ In: *Proceedings of the National Academy of Sciences* 73.3 (Mar. 1976), pp. 804–808. DOI: [10.1073/pnas.73.3.804](https://doi.org/10.1073/pnas.73.3.804). (Visited on 04/30/2026).
- [25] Itamar Sela and David B. Lukatsky. „DNA Sequence Correlations Shape Nonspecific Transcription Factor-DNA Binding Affinity“. In: *Biophysical Journal* 101.1 (July 2011), pp. 160–166. ISSN: 0006-3495, 1542-0086. DOI: [10.1016/j.bpj.2011.04.037](https://doi.org/10.1016/j.bpj.2011.04.037). (Visited on 05/04/2026).
- [26] Matthias Siebert and Johannes Söding. „Bayesian Markov Models Consistently Outperform PWMs at Predicting Motifs in Nucleotide Sequences“. In: *Nucleic Acids Research* 44.13 (July 2016), pp. 6055–6069. ISSN: 0305-1048. DOI: [10.1093/nar/gkw521](https://doi.org/10.1093/nar/gkw521). (Visited on 05/04/2026).
- [27] Matthew Slattery et al. „Absence of a Simple Code: How Transcription Factors Read the Genome“. In: *Trends in biochemical sciences* 39.9 (Sept. 2014), pp. 381–399. ISSN: 0968-0004. DOI: [10.1016/j.tibs.2014.07.002](https://doi.org/10.1016/j.tibs.2014.07.002). (Visited on 04/30/2026).
- [28] Can Sönmezer et al. „Molecular Co-Occupancy Identifies Transcription Factor Binding Cooperativity in Vivo“. In: *Molecular cell* 81.2 (Jan. 2021), 255–267.e6. ISSN: 1097-2765. DOI: [10.1016/j.molcel.2020.11.015](https://doi.org/10.1016/j.molcel.2020.11.015). (Visited on 05/08/2026).
- [29] Robert E. Thurman et al. „The Accessible Chromatin Landscape of the Human Genome“. In: *Nature* 489.7414 (Sept. 2012), pp. 75–82. ISSN: 1476-4687. DOI: [10.1038/nature11232](https://doi.org/10.1038/nature11232). (Visited on 05/04/2026).
- [30] Ilya E Vorontsov et al. „HOCOMOCO in 2024: A Rebuild of the Curated Collection of Binding Models for Human and Mouse Transcription Factors“. In: *Nucleic Acids Research* 52.D1 (Jan. 2024), pp. D154–D163. ISSN: 0305-1048. DOI: [10.1093/nar/gkad1077](https://doi.org/10.1093/nar/gkad1077). (Visited on 05/04/2026).
- [31] Matthew T. Weirauch et al. „Evaluation of Methods for Modeling Transcription Factor Sequence Specificity“. In: *Nature Biotechnology* 31.2 (Feb. 2013), pp. 126–134. ISSN: 1546-1696. DOI: [10.1038/nbt.2486](https://doi.org/10.1038/nbt.2486). (Visited on 05/04/2026).
- [32] Zeba Wunderlich and Leonid A. Mirny. „Different Gene Regulation Strategies Revealed by Analysis of Binding Motifs“. In: *Trends in Genetics* 25.10 (Oct. 2009), pp. 434–440. ISSN: 0168-9525. DOI: [10.1016/j.tig.2009.08.003](https://doi.org/10.1016/j.tig.2009.08.003). (Visited on 05/04/2026).
- [33] Lin Yang et al. „Transcription Factor Family-Specific DNA Shape Readout Revealed by Quantitative Specificity Models“. In: *Molecular Systems Biology* 13.2 (Feb. 2017), p. 910. ISSN: 1744-4292. DOI: [10.15252/msb.20167238](https://doi.org/10.15252/msb.20167238). (Visited on 04/30/2026).

Appendix

Sem můžete přidat přílohu. Pokud chcete “Přílohy”, tak upravte definici záhlaví v souboru sci.muni.thesis.sty, viz příkaz \HlavickaPriloha.

Quisque facilisis auctor sapien. Pellentesque gravida hendrerit lectus. Mauris rutrum sodales sapien. Fusce hendrerit sem vel lorem. Integer pellentesque massa vel augue. Integer elit tortor, feugiat quis, sagittis et, ornare non, lacus. Vestibulum posuere pellentesque eros. Quisque venenatis ipsum dictum nulla. Aliquam quis quam non metus eleifend interdum. Nam eget sapien ac mauris malesuada adipiscing. Etiam eleifend neque sed quam. Nulla facilisi. Proin a ligula. Sed id dui eu nibh egestas tincidunt. Suspendisse arcu.

Maecenas dui. Aliquam volutpat auctor lorem. Cras placerat est vitae lectus. Curabitur massa lectus, rutrum euismod, dignissim ut, dapibus a, odio. Ut eros erat, vulputate ut, interdum non, porta eu, erat. Cras fermentum, felis in porta congue, velit leo facilisis odio, vitae consectetur lorem quam vitae orci. Sed ultrices, pede eu placerat auctor, ante ligula rutrum tellus, vel posuere nibh lacus nec nibh. Maecenas laoreet dolor at enim. Donec molestie dolor nec metus. Vestibulum libero. Sed quis erat. Sed tristique. Duis pede leo, fermentum quis, consectetur eget, vulputate sit amet, erat.

Donec vitae velit. Suspendisse porta fermentum mauris. Ut vel nunc non mauris pharetra varius. Duis consequat libero quis urna. Maecenas at ante. Vivamus varius, wisi sed egestas tristique, odio wisi luctus nulla, lobortis dictum dolor ligula in lacus. Vivamus aliquam, urna sed interdum porttitor, metus orci interdum odio, sit amet euismod lectus felis et leo. Praesent ac wisi. Nam suscipit vestibulum sem. Praesent eu ipsum vitae pede cursus venenatis. Duis sed odio. Vestibulum eleifend. Nulla ut massa. Proin rutrum mattis sapien. Curabitur dictum gravida ante.

Phasellus placerat vulputate quam. Maecenas at tellus. Pellentesque neque diam, dignissim ac, venenatis vitae, consequat ut, lacus. Nam nibh. Vestibulum fringilla arcu mollis arcu. Sed et turpis. Donec sem tellus, volutpat et, varius eu, commodo sed, lectus. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Quisque enim arcu, suscipit nec, tempus at, imperdiet vel, metus. Morbi volutpat purus at erat. Donec dignissim, sem id semper tempus, nibh massa eleifend turpis, sed pellentesque wisi purus sed libero. Nullam lobortis tortor vel risus. Pellentesque consequat nulla eu tellus. Donec velit. Aliquam fermentum, wisi ac rhoncus iaculis, tellus nunc malesuada orci, quis volutpat dui magna id mi. Nunc vel ante. Duis vitae lacus. Cras nec ipsum.

Morbi nunc. Aliquam consectetur varius nulla. Phasellus eros. Cras dapibus porttitor risus. Maecenas ultrices mi sed diam. Praesent gravida velit at elit vehicula porttitor. Phasellus nisl mi, sagittis ac, pulvinar id, gravida sit amet, erat. Vestibulum est. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Curabitur id sem elementum leo ru-

trum hendrerit. Ut at mi. Donec tincidunt faucibus massa. Sed turpis quam, sollicitudin a, hendrerit eget, pretium ut, nisl. Duis hendrerit ligula. Nunc pulvinar congue urna.

Nunc velit. Nullam elit sapien, eleifend eu, commodo nec, semper sit amet, elit. Nulla lectus risus, condimentum ut, laoreet eget, viverra nec, odio. Proin lobortis. Curabitur dictum arcu vel wisi. Cras id nulla venenatis tortor congue ultrices. Pellentesque eget pede. Sed eleifend sagittis elit. Nam sed tellus sit amet lectus ullamcorper tristique. Mauris enim sem, tristique eu, accumsan at, scelerisque vulputate, neque. Quisque lacus. Donec et ipsum sit amet elit nonummy aliquet. Sed viverra nisl at sem. Nam diam. Mauris ut dolor. Curabitur ornare tortor cursus velit.

Morbi tincidunt posuere arcu. Cras venenatis est vitae dolor. Vivamus scelerisque semper mi. Donec ipsum arcu, consequat scelerisque, viverra id, dictum at, metus. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut pede sem, tempus ut, porttitor bibendum, molestie eu, elit. Suspendisse potenti. Sed id lectus sit amet purus faucibus vehicula. Praesent sed sem non dui pharetra interdum. Nam viverra ultrices magna.

Aenean laoreet aliquam orci. Nunc interdum elementum urna. Quisque erat. Nullam tempor neque. Maecenas velit nibh, scelerisque a, consequat ut, viverra in, enim. Duis magna. Donec odio neque, tristique et, tincidunt eu, rhoncus ac, nunc. Mauris malesuada malesuada elit. Etiam lacus mauris, pretium vel, blandit in, ultricies id, libero. Phasellus bibendum erat ut diam. In congue imperdiet lectus.

Aenean scelerisque. Fusce pretium porttitor lorem. In hac habitasse platea dictumst. Nulla sit amet nisl at sapien egestas pretium. Nunc non tellus. Vivamus aliquet. Nam adipiscing euismod dolor. Aliquam erat volutpat. Nulla ut ipsum. Quisque tincidunt auctor augue. Nunc imperdiet ipsum eget elit. Aliquam quam leo, consectetur non, ornare sit amet, tristique quis, felis. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Pellentesque interdum quam sit amet mi. Pellentesque mauris dui, dictum a, adipiscing ac, fermentum sit amet, lorem.

Ut quis wisi. Praesent quis massa. Vivamus egestas risus eget lacus. Nunc tincidunt, risus quis bibendum facilisis, lorem purus rutrum neque, nec porta tortor urna quis orci. Aenean aliquet, libero semper volutpat luctus, pede erat lacinia augue, quis rutrum sem ipsum sit amet pede. Vestibulum aliquet, nibh sed iaculis sagittis, odio dolor blandit augue, eget mollis urna tellus id tellus. Aenean aliquet aliquam nunc. Nulla ultricies justo eget orci. Phasellus tristique fermentum leo. Sed massa metus, sagittis ut, semper ut, pharetra vel, erat. Aliquam quam turpis, egestas vel, elementum in, egestas sit amet, lorem. Duis convallis, wisi sit amet mollis molestie, libero mauris porta dui, vitae aliquam arcu turpis ac sem. Aliquam aliquet dapibus metus.

Vivamus commodo eros eleifend dui. Vestibulum in leo eu erat tristique mattis. Cras at elit. Cras pellentesque. Nullam id lacus sit amet libero aliquet hendrerit. Proin placerat, mi non elementum laoreet, eros elit tincidunt magna, a rhoncus sem arcu id odio. Nulla eget leo a leo egestas facilisis. Curabitur quis velit. Phasellus aliquam, tortor nec ornare rhoncus, purus urna posuere velit, et commodo risus tellus quis tellus. Vivamus leo turpis, tempus sit amet, tristique vitae, laoreet quis, odio. Proin scelerisque bibendum ipsum. Etiam nisl. Praesent vel dolor. Pellentesque vel magna. Curabitur urna. Vivamus congue urna in velit. Etiam ullamcorper elementum dui. Praesent non urna. Sed placerat quam non mi. Pellentesque diam magna, ultricies eget, ultrices placerat, adipiscing rutrum, sem.

Morbi sem. Nulla facilisi. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Nulla facilisi. Morbi sagittis ultrices libero. Praesent eu ligula sed sapien auctor sagittis. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Donec vel nunc. Nunc fermentum, lacus id aliquam porta, dui tortor euismod eros, vel molestie ipsum purus eu lacus. Vivamus pede arcu, euismod ac, tempus id, pretium et, lacus. Curabitur sodales dapibus urna. Nunc eu sapien. Donec eget nunc a pede dictum pretium. Proin mauris. Vivamus luctus libero vel nibh.

Fusce tristique risus id wisi. Integer molestie massa id sem. Vestibulum vel dolor. Pellentesque vel urna vel risus ultricies elementum. Quisque sapien urna, blandit nec, iaculis ac, viverra in, odio. In hac habitasse platea dictumst. Morbi neque lacus, convallis vitae, commodo ac, fermentum eu, velit. Sed in orci. In fringilla turpis non arcu. Donec in ante. Phasellus tempor feugiat velit. Aenean varius massa non turpis. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae;

Aliquam tortor. Morbi ipsum massa, imperdiet non, consectetur vel, feugiat vel, lorem. Quisque eget lorem nec elit malesuada vestibulum. Quisque sollicitudin ipsum vel sem. Nulla enim. Proin nonummy felis vitae felis. Nullam pellentesque. Duis rutrum feugiat felis. Mauris vel pede sed libero tincidunt mollis. Phasellus sed urna rhoncus diam euismod bibendum. Phasellus sed nisl. Integer condimentum justo id orci iaculis varius. Quisque et lacus. Phasellus elementum, justo at dignissim auctor, wisi odio lobortis arcu, sed sollicitudin felis felis eu neque. Praesent at lacus.

Vivamus sit amet pede. Duis interdum, nunc eget rutrum dignissim, nisl diam luctus leo, et tincidunt velit nisl id tellus. In lorem tellus, aliquet vitae, porta in, aliquet sed, lectus. Phasellus sodales. Ut varius scelerisque erat. In vel nibh eu eros imperdiet rutrum. Donec ac odio nec neque vulputate suscipit. Nam nec magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Nullam porta, odio et sagittis iaculis, wisi neque fringilla sapien, vel commodo lorem lorem id elit. Ut sem lectus, scelerisque eget, placerat et, tincidunt scelerisque, ligula. Pellentesque non orci.

Etiam vel ipsum. Morbi facilisis vestibulum nisl. Praesent cursus laoreet felis. Integer adipiscing pretium orci. Nulla facilisi. Quisque posuere bibendum purus. Nulla quam mauris, cursus eget, convallis ac, molestie non, enim. Aliquam congue. Quisque sagittis nonummy sapien. Proin molestie sem vitae urna. Maecenas lorem. Vivamus viverra consequat enim.

